Economic Transformation

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Summary

Most FGD groups reported that CFM had not improved access to natural resources for either household use or sale, though approximately one third indicated that access had increased. Community youth were significantly less likely than CFMG executive FGDs to report improved access for household use. Individual interviewees showed a similar pattern, with DFOs significantly more likely than CFMG executives or traditional leaders to consider that CFM had improved household resource access. Employment outcomes were more positive. Approximately half of FGD groups agreed that CFM had generated new employment opportunities, with a marked contrast between community youth (who were most likely to agree) and community female FGDs (who were most likely to disagree). Across the surveyed CFMGs, reported employment numbers increased substantially after CFM establishment, though there was considerable variation among provinces (means ranging from 3 in Western Province to 21 in Lusaka Province). Employment was strongly male-biased and the dominant employment category was HFO or Community Scout. Income earning opportunities from CFM remain limited for most communities. Around one third of CFMGs had received carbon finance payments, and awareness of carbon credit agreements was generally high. Carbon trader support was primarily directed at improved forest management and agricultural extension, rather than at developing forest product value chains. However, forest product value chains — particularly honey and mushrooms — dominated both realised and aspirational income sources among CF communities. Most forest products are sold only at local markets and private enterprise partnerships were uncommon, with the vast majority involving honey off-takers. Communities identified timber as the most important unrealised commercial opportunity, despite none of the surveyed CFMGs being involved in timber production. Opportunity costs of CFM were recognised by only about one third of FGD groups. Unexpectedly, non-timber forest product (NTFP) harvesting was frequently cited as a precluded activity, suggesting possible misunderstandings around forest use rules in some communities. It is evident from the survey that most CFMG business plans are overly reliant on carbon and potentially valuable NTFP value chains remain under-exploited.

Introduction

In this chapter, we consider the effect of CFM management on indicators of economic transformation.

Access to natural resources

We asked FGD groups whether CFM had improved access to natural resources for household use or for sale (Figure 1, Figure 2).

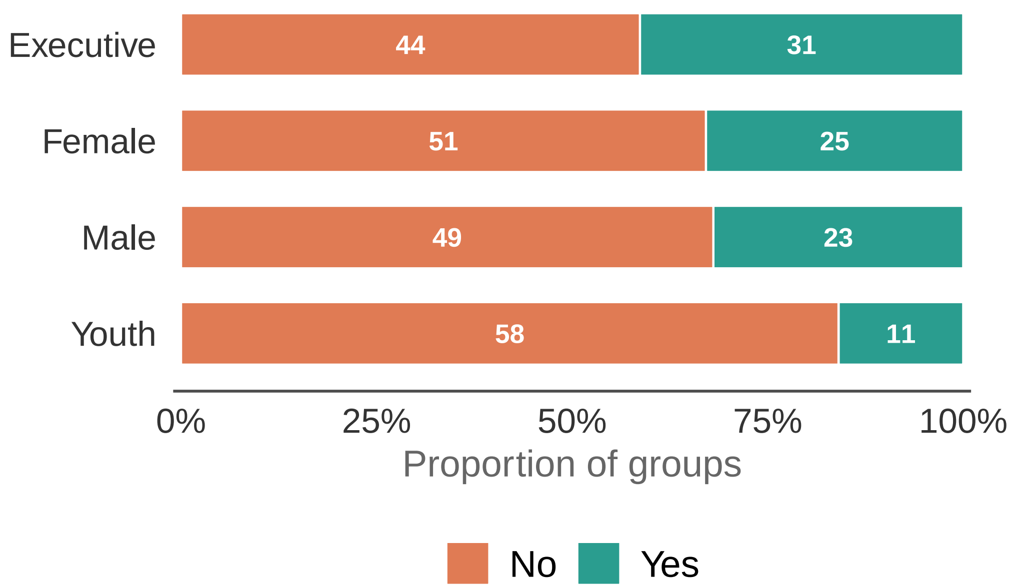

Figure 1: Has CFM improved access to natural resources for household use? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Responses were similar for both household use and sale. Most FGD groups said that there was no increase in access to natural resources for household use or for sale. However, approximately one third of the groups indicated that access to natural resources has increased under CFM. Community youth were significantly more likely to say there had not been any improvement in access to natural resources for household use, as compared to CFMG executive FGDs. None of the other pairwise comparisons were significant.

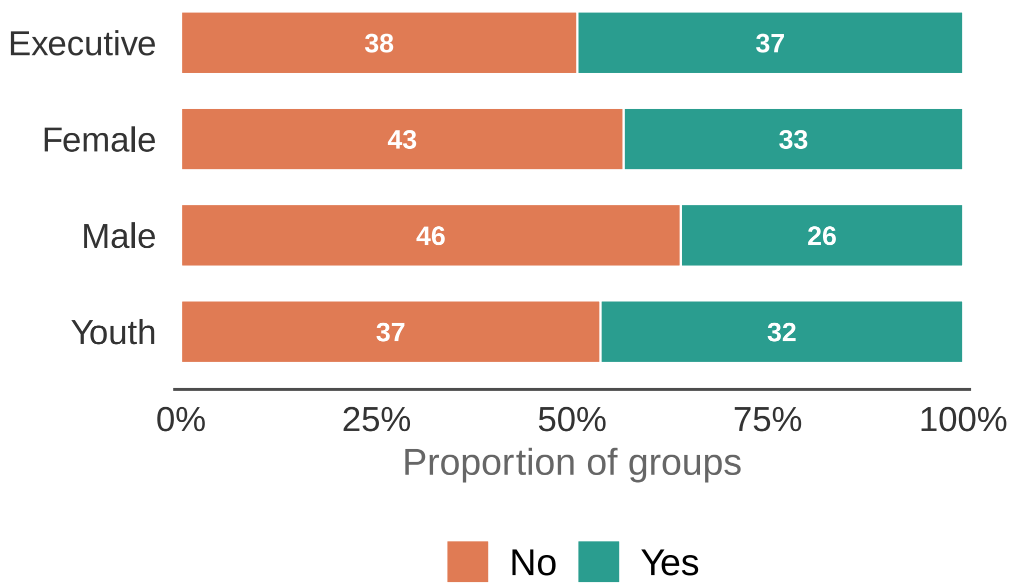

Figure 2: Has CFM improved access to natural resources for sale? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Table 1: Results of the model examining the effect of FGD type and governance index on whether communities report improved access to natural resources for household use.

Model results for improved access to natural resources for household use <U+2014> FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0.117; CF-within-province variance = 0.327. Marginal R<U+00B2> = 0.087; conditional R<U+00B2> = 0.196. n = 273.

Factor	Estimate	SE	*P* value
Intercept (Executive FGD)	−1.060	0.461	0.0214
Female FGD	−0.433	0.370	0.2429
Male FGD	−0.471	0.379	0.2138
Youth FGD	−1.379	0.430	0.0013
Governance index	0.145	0.078	0.0633

Table 2: Pairwise comparisons among FGD types for improved access to natural resources for household use.

Pairwise comparisons <U+2014> FGD type

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	*P* value
Executive - Female	0.433	0.370	1.168	0.6472
Executive - Male	0.471	0.379	1.243	0.5993
Executive - Youth	1.379	0.430	3.209	0.0073
Female - Male	0.039	0.394	0.098	0.9997
Female - Youth	0.946	0.439	2.156	0.1359
Male - Youth	0.908	0.446	2.033	0.1757

Table 3: Results of the model examining the effect of FGD type and governance index on whether communities report improved access to natural resources for sale.

Model results for improved access to natural resources for sale <U+2014> FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0.772; CF-within-province variance = 1.542. Marginal R<U+00B2> = 0.054; conditional R<U+00B2> = 0.445. n = 273.

Factor	Estimate	SE	*P* value
Intercept (Executive FGD)	−1.281	0.664	0.0537
Female FGD	−0.161	0.401	0.6881

Male FGD	−0.561	0.415	0.1761
Youth FGD	−0.064	0.404	0.8744
Governance index	0.246	0.109	0.0239

Individual interviewees were also asked whether CFM had improved access to natural resources for household use or for sale. Table 4 shows the percentage reporting each by interviewee role. Again a majority of respondents indicated that they did not think that CFM had improved access to natural resources for either household use or sale. DFOs were significantly more likely to suggest that CFM had improved access to natural resources for household use than CFMG executives or traditional leaders. None of the other pairwise comparisons were significant.

Table 4: Percentage of individual interviewees reporting improved access to natural resources as an economic benefit of CFM.

Improved access to natural resources from CFM as reported by individual interviewees

Values represent the % of responses agreeing that CFM has improved assess to resources for household use or sale. n = the total number of interview responses.

Role	n	Household use	For sale
Community member	31	45%	52%
CFMG executive member	63	24%	33%
Traditional leader	75	27%	39%
District Forestry Officer	56	62%	64%
NGO representative	32	50%	38%

Table 5: Results of the model examining the effect of interviewee role, gender and age on whether individuals report improved access to natural resources for household use.

Model results for improved access to natural resources for household use <U+2014> individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0.129; CF-within-province variance = 0.149. Marginal R<U+00B2> = 0.122; conditional R<U+00B2> = 0.19. n = 257.

Factor	Estimate	SE	P value
Intercept (Community member)	0.235	0.810	0.7717
CFMG executive member	−1.023	0.527	0.0520
Traditional leader	−0.823	0.490	0.0934
District Forestry Officer	0.695	0.502	0.1663
NGO representative	0.098	0.555	0.8599
Male	−0.317	0.372	0.3943

Age	−0.002	0.013	0.8479

Table 6: Pairwise comparisons among interviewee roles for improved access to natural resources for household use.

Pairwise comparisons <U+2014> interviewee role

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	*P* value
cfmg_member - cfmg_exec_member	1.023	0.527	1.943	0.2946
cfmg_member - trad_leader	0.823	0.490	1.678	0.4476
cfmg_member - dfo	−0.695	0.502	−1.384	0.6380
cfmg_member - NGO_rep	−0.098	0.555	−0.177	0.9998
cfmg_exec_member - trad_leader	−0.200	0.460	−0.435	0.9926
cfmg_exec_member - dfo	−1.718	0.474	−3.628	0.0026
cfmg_exec_member - NGO_rep	−1.121	0.515	−2.175	0.1892
trad_leader - dfo	−1.518	0.453	−3.347	0.0073
trad_leader - NGO_rep	−0.921	0.515	−1.788	0.3807
dfo - NGO_rep	0.597	0.480	1.245	0.7250

Table 7: Results of the model examining the effect of interviewee role, gender and age on whether individuals report improved access to natural resources for sale.

Model results for improved access to natural resources for sale <U+2014> individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0.143; CF-within-province variance = 0.333. Marginal R<U+00B2> = 0.073; conditional R<U+00B2> = 0.19. n = 257.

Factor	Estimate	SE	*P* value
Intercept (Community member)	0.998	0.817	0.2218
CFMG executive member	−0.623	0.519	0.2293
Traditional leader	−0.464	0.492	0.3463
District Forestry Officer	0.493	0.519	0.3425
NGO representative	−0.765	0.576	0.1843
Male	−0.448	0.365	0.2198
Age	−0.012	0.013	0.3449

Employment

We asked FGD groups whether CFM had generated new employment opportunities (Figure 3).

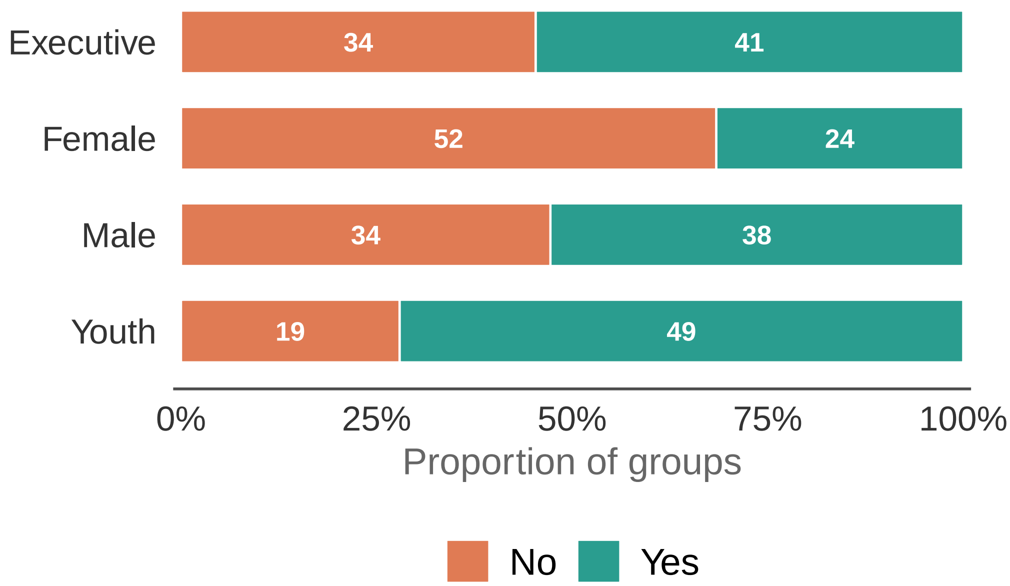

Figure 3: Responses to whether CFM has generated employment opportunities. The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

As can be seen from the figure, approximately half of the FGD groups agreed that CFM had increased employment opportunities. Interestingly, community youth were especially likely to agreed that CFM had created employment opportunities. In contrast, community female FGD groups were much more likely to say that CFM had not increased employment opportunities. All pairwise comparisons were highly significant.

Table 8: Results of the model examining the effect of FGD type and governance index on whether communities report that CFM has generated employment opportunities.

Model results for employment opportunities generated by CFM <U+2014> FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0.893; CF-within-province variance = 0. Marginal R<U+00B2> = 0.287; conditional R<U+00B2> = 0.439. n = 272.

Factor	Estimate	SE	P value
Intercept (Executive FGD)	−1.617	0.566	0.0043
Female FGD	−1.459	0.419	0.0005
Male FGD	−0.384	0.400	0.3373
Youth FGD	1.124	0.427	0.0085
Governance index	0.434	0.082	0.0000

Table 9: Pairwise comparisons among FGD types for whether CFM has generated employment opportunities.

Pairwise comparisons <U+2014> FGD type

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	P value
Executive - Female	1.459	0.419	3.484	0.0028
Executive - Male	0.384	0.400	0.959	0.7725
Executive - Youth	−1.124	0.427	−2.633	0.0421
Female - Male	−1.076	0.429	−2.508	0.0587
Female - Youth	−2.583	0.473	−5.457	0.0000
Male - Youth	−1.508	0.445	−3.387	0.0039

Individual interviewees were also asked whether CFM had generated employment opportunities (Table 10). Interestingly, community respondents were much more likely to agree that CFM had created employment opportunities than NGO representatives or DFOs. The differences between DFOs and CFMG executives and traditional leaders were significant, but none of the other pairwise comparisons were significant.

Table 10: Percentage of individual interviewees reporting that CFM has generated employment opportunities by interviewee role.

Employment opportunities reported by individual interviewees

Values represent the % of responses agreeing that CFM has generated employment opportunities. n = the total number of interview responses.

Role	n	Agree CFM has generated employment opportunities
Community member	31	60%
CFMG executive member	63	75%
Traditional leader	75	53%
District Forestry Officer	56	38%
NGO representative	32	48%

Table 11: Results of the model examining the effect of interviewee role, gender and age on whether individuals report that CFM has generated employment opportunities.

Model results for employment opportunities generated by CFM <U+2014> individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0.314; CF-within-province variance = 0.195. Marginal R<U+00B2> = 0.116; conditional R<U+00B2> = 0.235. n = 250.

Factor	Estimate	SE	*P* value
Intercept (Community member)	2.211	0.922	0.0165
CFMG executive member	0.331	0.564	0.5569
Traditional leader	−0.088	0.531	0.8686
District Forestry Officer	−1.448	0.580	0.0125
NGO representative	−0.975	0.634	0.1241
Male	0.436	0.380	0.2513
Age	−0.041	0.014	0.0031

Table 12: Pairwise comparisons among interviewee roles for whether CFM has generated employment opportunities.

Pairwise comparisons <U+2014> interviewee role

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	*P* value
cfmg_member - cfmg_exec_member	−0.331	0.564	−0.587	0.9770
cfmg_member - trad_leader	0.088	0.531	0.165	0.9998
cfmg_member - dfo	1.448	0.580	2.498	0.0909
cfmg_member - NGO_rep	0.975	0.634	1.538	0.5377
cfmg_exec_member - trad_leader	0.419	0.463	0.905	0.8953
cfmg_exec_member - dfo	1.779	0.484	3.677	0.0022
cfmg_exec_member - NGO_rep	1.307	0.552	2.367	0.1245
trad_leader - dfo	1.360	0.471	2.890	0.0315
trad_leader - NGO_rep	0.887	0.543	1.633	0.4761
dfo - NGO_rep	−0.473	0.502	−0.941	0.8809

Changes in numbers employed

We asked CFMG Executive FGDs to estimate the number of people employed before and after CFM (Figure 4). The mean number of people employed per CFMG varied from three in Western Province to 21 in Lusaka Province. The figure also indicates that the numbers of people employed in forest management had substantially increased.

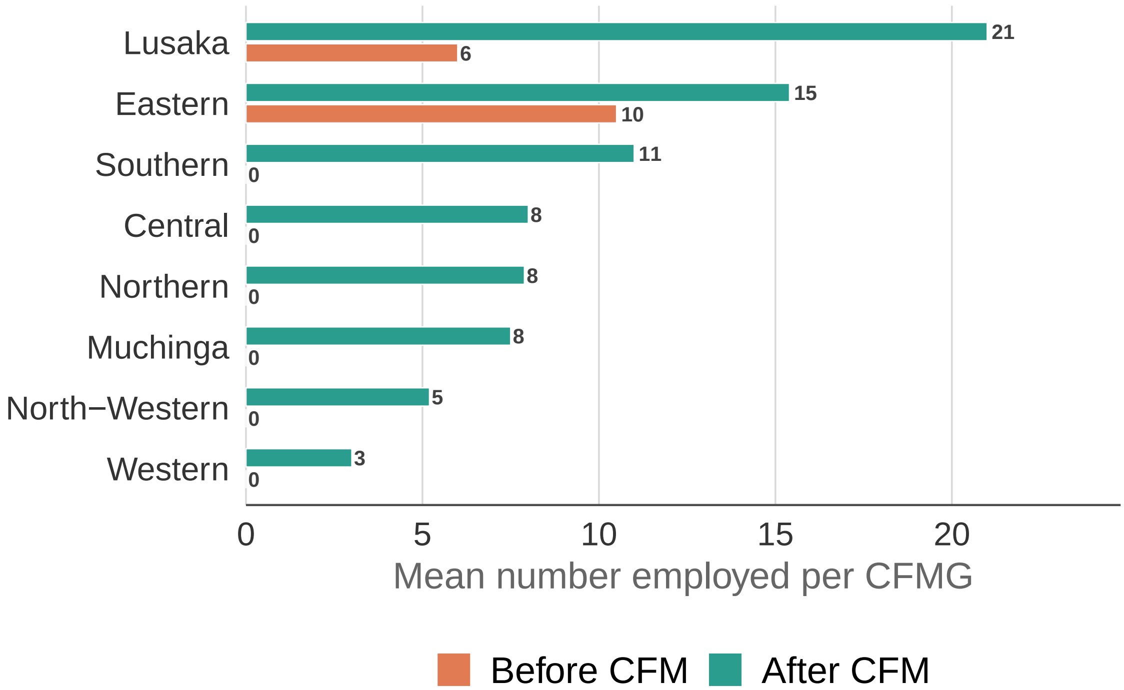

Figure 4: Mean number of people per CFMG employed before and after CFM by province, as reported by CFMG Executives.

Full-time senior employment by gender and province

CFMG Executive FGDs also reported on the number of senior males and females in full-time employment in the CFMG (Figure 5). Clearly, there are many more senior men in full-time employment than women. According to the response across the 75 CFMGs surveyed, 278 senior men were employed as compared to 56 women.

CFMG Executive FGDs also reported the number of male and female youth in full-time employment in the CFMG (Figure 6). Across the sample of 75 CFMGs, 207 male youths were employed compared to 48 female youths.

Types of formal employment opportunities

Community female, male, and youth FGDs identified the types of formal employment opportunities generated by CFM (Figure 7). Clearly, by far the largest type of job created is HFO or Community Scout.

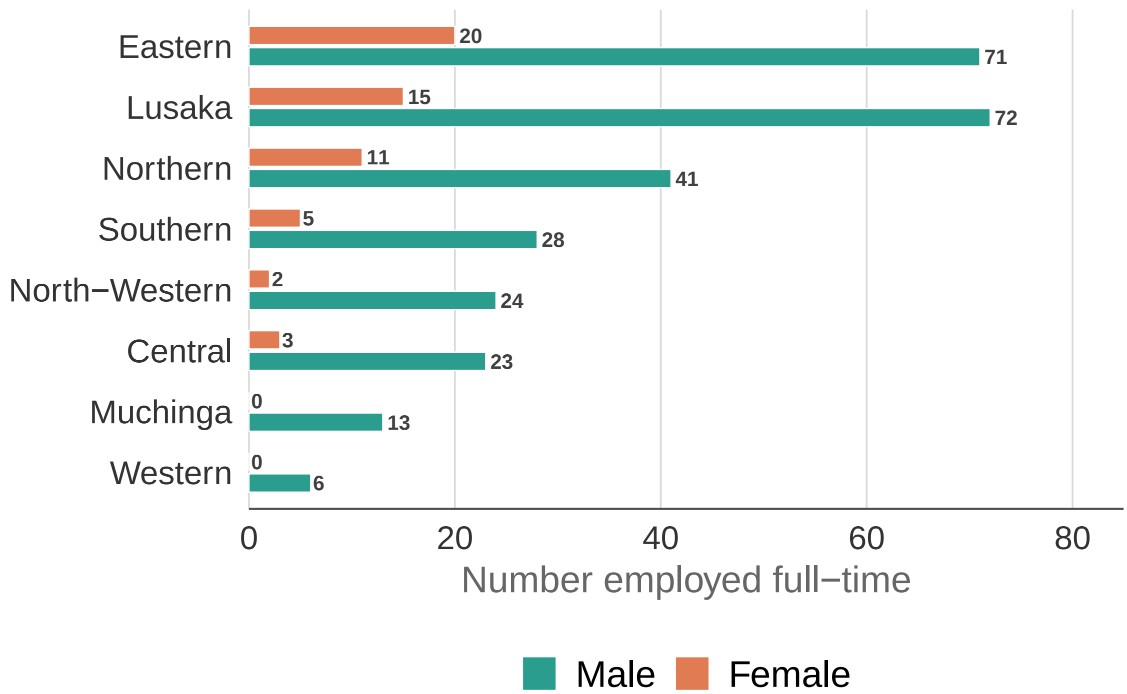

Figure 5: Number of males and females employed full-time in CFMGs, by province, as reported by executive FGDs.

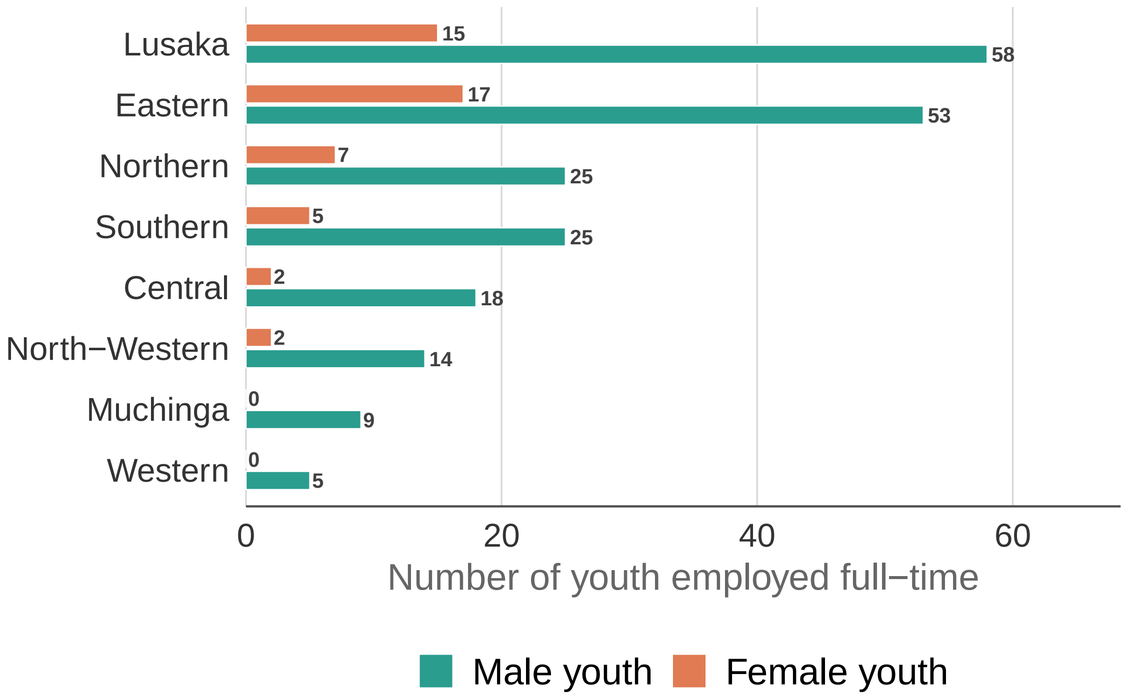

Figure 6: Number of male and female youth employed full-time in CFMGs, by province, as reported by executive FGDs.

Figure 7: Types of formal employment opportunities reported by female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting that job type.

Income earning opportunities

Carbon finance

We asked the CFMG executive FGDs whether their CFMG had received income from carbon finance (Figure 8). According to the responses, around one third of CFs have received carbon finance. Later responses suggest this is an under-representation for the number of CFMGs involved in carbon finance and likely reflects only those CFMGs that have actually started to receive payouts from carbon.

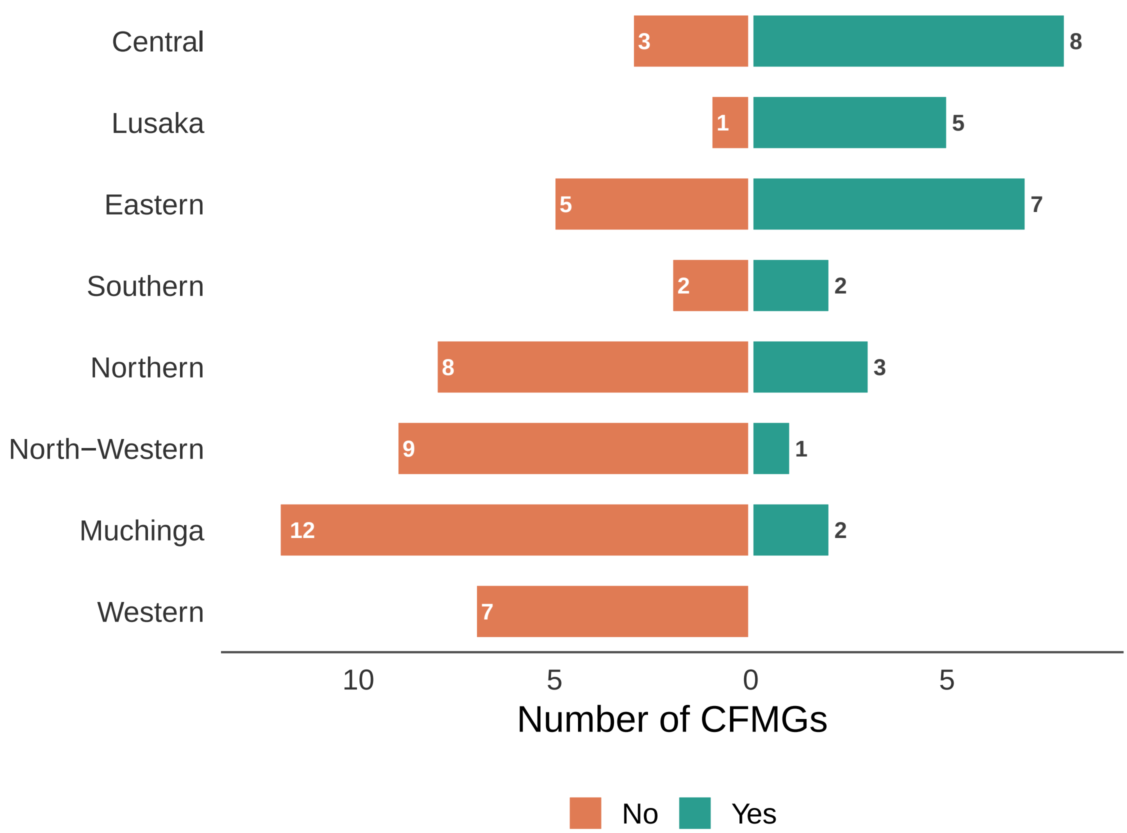

Figure 8: CFMG involvement in carbon finance by province, as reported by CFMG executives.

FGD groups were also asked whether the community was aware of the carbon credit agreement (Figure 9). While most CFMG executive groups stated that the community was aware of the carbon finance agreement, interestingly two stated that they were not, which is a little surprising. Most community female and male groups indicated they were aware of the carbon finance agreement, but more than half of the community youth groups stated they were not aware of the carbon finance agreement. Community youth were significantly less likely to know about the carbon finance agreement than CFMG Executives, but other pairwise comparisons were not significant.

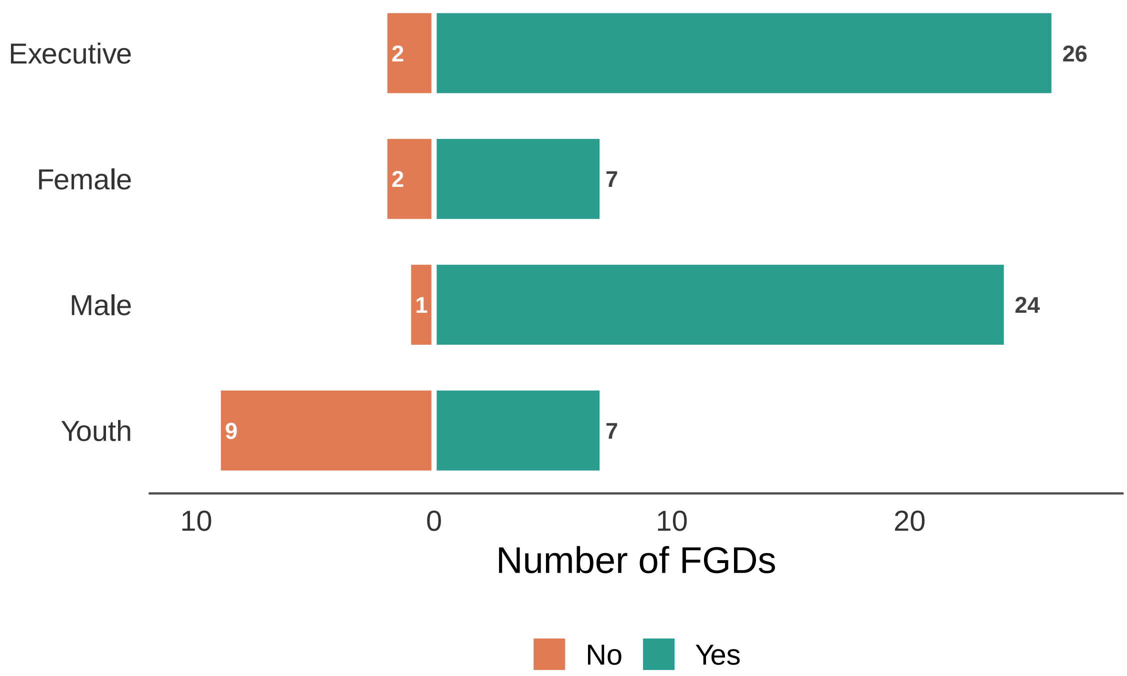

Figure 9: Community awareness of the carbon credit agreement by FGD group.

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Random effect variances not available. Returned R2 does not account for random effects.
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Table 13: Results of the model examining the effect of FGD type and governance index on community awareness of the carbon credit agreement.

Model results for community awareness of carbon credit agreement <U+2014> FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0; CF-within-province variance = 0. Marginal R<U+00B2> = 0.988; conditional R<U+00B2> = NA. n = 70.

Factor	Estimate	SE	P value
Intercept (Executive FGD)	3.499	1.214	0.0039
Female FGD	−1.681	1.127	0.1359
Male FGD	34.489	15,005,998.180	1.0000
Youth FGD	−2.659	0.907	0.0034

Governance index	−0.189	0.184	0.3051

Table 14: Pairwise comparisons among FGD types for community awareness of the carbon credit agreement.

Pairwise comparisons <U+2014> FGD type

Estimates are log-odds differences on the emmeans scale. Tukey-adjusted p-values.

Contrast	Estimate	SE	z ratio	*P* value
Executive - Female	1.681	1.127	1.491	0.4428
Executive - Male	−34.489	15,005,998.180	0.000	1.0000
Executive - Youth	2.659	0.907	2.932	0.0177
Female - Male	−36.169	15,005,998.180	0.000	1.0000
Female - Youth	0.979	1.002	0.976	0.7629
Male - Youth	37.148	15,005,998.180	0.000	1.0000

Individual interviewees were also asked whether the community was aware of the carbon credit agreement (Table 15). Most respondents indicated that the community was aware, but the percentage of positive responses varied across interviewee roles. Interestingly, the NGO represenatatives (a groups that also included carbon enterprise representatives), overwhelmingly (~95%) suggested the community was informed, but the only 70% of DFOs thought the communities were informed. However, none of these differences were significant.

Table 15: Percentage of individual interviewees reporting community awareness of the carbon credit agreement, by interviewee role.

Community awareness of the carbon credit agreement, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses.

Role	n	Aware of carbon credit agreement
Community member	31	83%
CFMG executive member	63	87%
Traditional leader	75	80%
District Forestry Officer	56	70%
NGO representative	32	95%

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Random effect variances not available. Returned R2 does not account for random effects.
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Table 16: Results of the model examining the effect of interviewee role, gender and age on community awareness of the carbon credit agreement.

Model results for community awareness of carbon credit agreement <U+2014> individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0; CF-within-province variance = 0. Marginal R<U+00B2> = 0.94; conditional R<U+00B2> = NA. n = 97.

Factor	Estimate	SE	*P* value
Intercept (Community member)	19.333	6,639.258	0.9977
CFMG executive member	0.305	0.981	0.7554
Traditional leader	−0.038	1.070	0.9716
District Forestry Officer	−0.456	0.985	0.6435
NGO representative	1.645	1.336	0.2183
Male	−19.029	6,639.258	0.9977
Age	0.020	0.029	0.4943

Carbon trader support

The types of CFM activities supported by the carbon trader are shown in Figure 10. As can be seen support focused on aspects of forest management or agricultural extension. Other potential elements of support, such as NTFPs, infrastructure development or market access, featured substantially less.

Other income earning opportunities

FGD groups also reported whether CFM had generated income earning opportunities through specific interventions, including market linkages, alternative livelihood interventions, conservation revenue, forest product processing and ecotourism (Figure 11). Although some community FGDs indicated improvement in income earning opportunities under some categories, especially conservation revenue and forest product processing, most indicated that CFM had not generated new income earning opportunities in these areas.

However, community female, male, and youth FGD groups did identified other additional sources of income generated by CFM (Figure 12). Natural resource based value chains featured prominently among these, but interestingly community business was also mentioned.

To better understand the importance of different income earning opportunities we asked the community female, male, and youth FGD groups to rank their top three additional sources of income from CFM (Figure 13). Interestingly, Among the 14 value chains identified, 12 were mentioned as 1st rank at least once. This immediately points to the diversity of forest products being harvested from CFs across the country, but also to the fact that for most CFs

Figure 10: Types of CFM activities supported by the carbon trader.

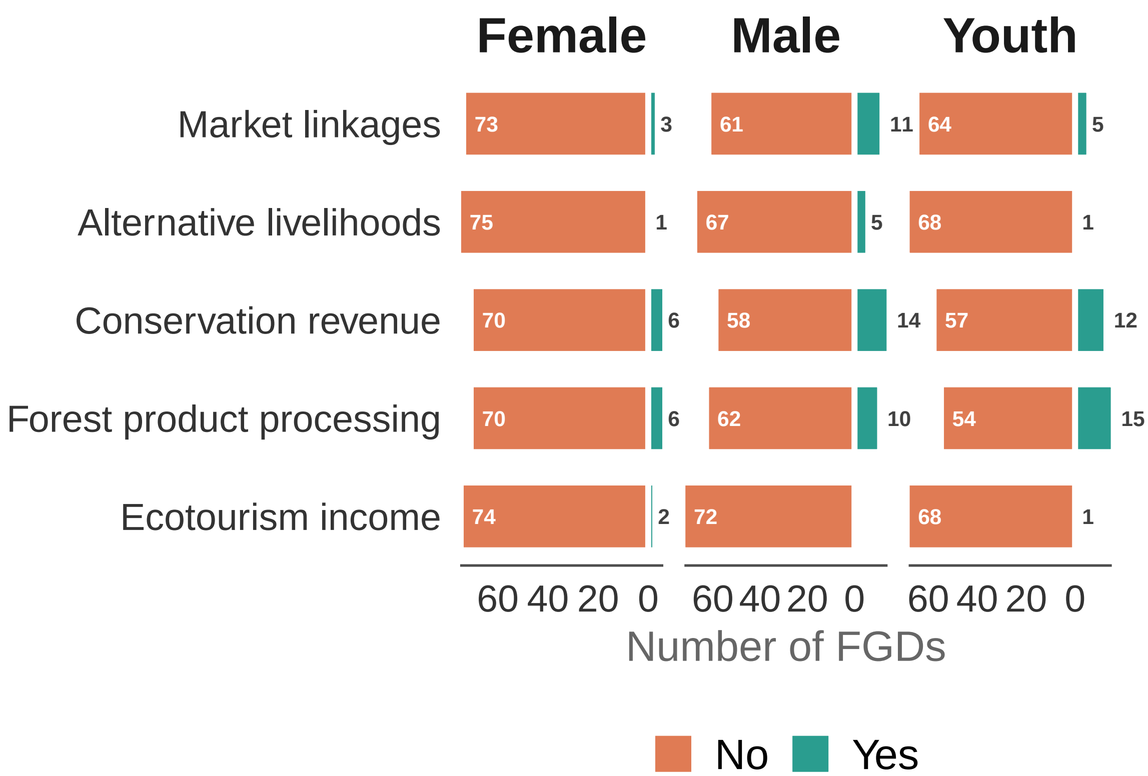

Figure 11: Income earning opportunities generated by CFM, as reported by female, male, and youth FGDs. The length of the bars represents the number of FGDs reporting a benefit (Yes) or no benefit (No) for each income source.

Figure 12: Additional sources of income from CFM reported by female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting each income source.

only one product features strongly as an addition source of income. Among these, mushrooms and honey are clearly the most frequently mentioned.

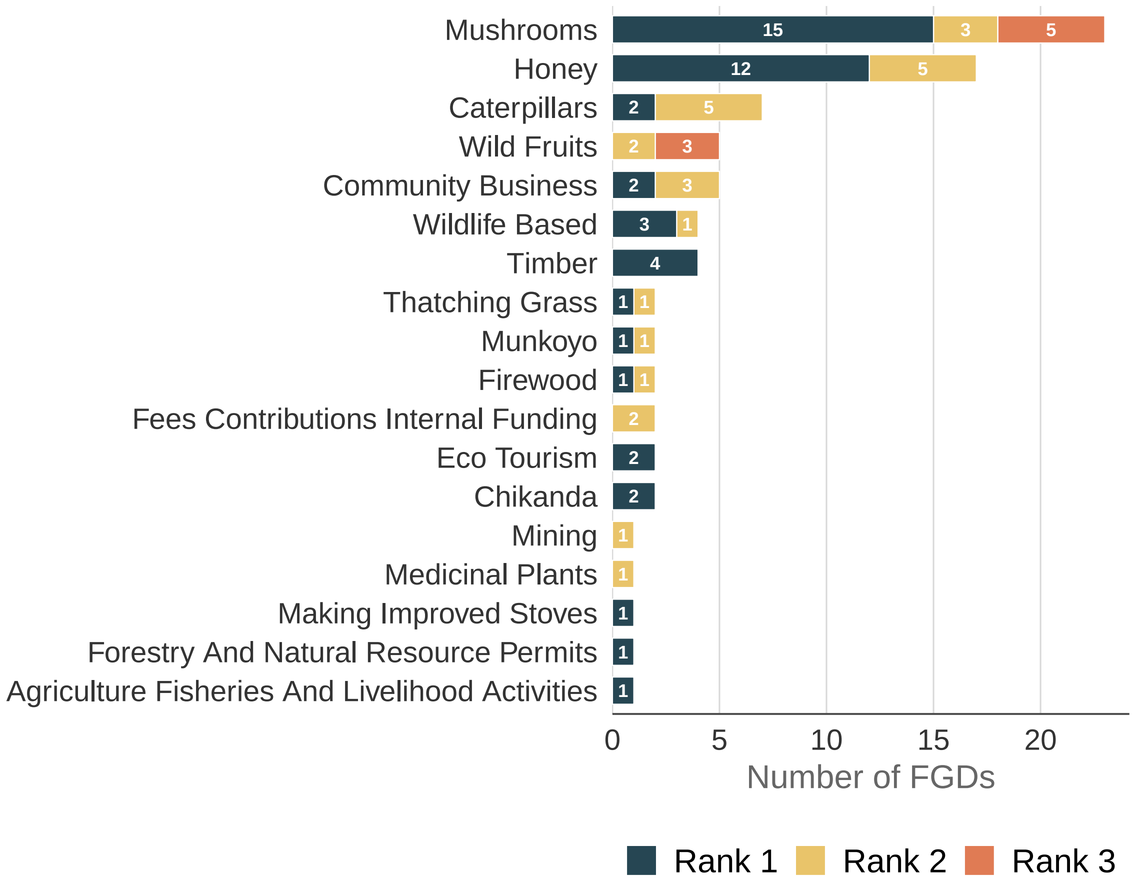

Figure 13: Ranking of additional income sources from CFM by female, male, and youth FGDs. Bars show the number of FGDs assigning each rank. Only income sources receiving at least one top-three ranking are shown.

CFMG executive FGDs were asked the estimated annual income generated through CFM (Figure 14). Unfortunately, there was a lot of inconsistency in the responses to this question. Most FGD groups did not include income from carbon credit sales, but some did. The figure below includes only those that did not include the carbon credit income to facilitate comparison.

Female, male, and youth FGDs also reported which forest products were harvested for household use (Figure 15). Mushrooms again featured prominently, but other natural resources are also clearly important, such as wild fruits, medicinal plants, thatching grass and caterpillars.

CFMG executive FGDs were asked whether the CFMG was collaborating with any private enterprises (Figure 16). The answers indicate that only a small number of CFMGs were

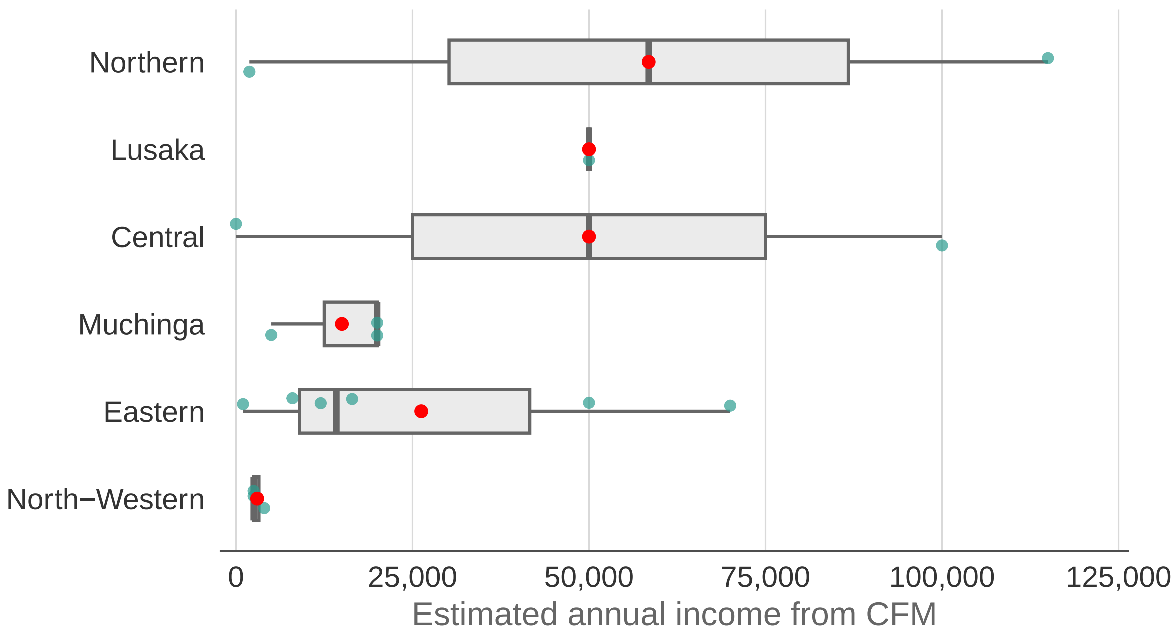

Figure 14: Mean estimated annual income from CFM by province, as reported by executive FGDs.

collaborating with a private enterprise.

The value chains that CFMGs reported working with through private enterprise partnerships are shown in Figure 17. Sixteen out of 17 involved partnerships with honey off-takers and just one involved mushrooms.

We asked the community female, male, and youth FGD groups to identify economic opportunities that remained unrealised within their CFs (Figure 18). Interestingly, although a wide range of different natural resource related value chains were mentioned, timber was mention the most often. This is noteworthy as none of the CFMGs we surveyed are involved in timber production.

The ranking of these unrealised opportunities is shown in Figure 19. Again a wide range of resources are mentioned, but the top three are timber, honey and mushrooms. Again, it is interesting that community groups consider their timber resources to have high unrealised potential for commercial development.

FGD groups also identified potential value-added opportunities from CFM resources (Figure 20). The overlap with the previous two questions is substantial, but it is interesting that the communities specifically identify honey processing and drying mushrooms, as well as timber processing. This suggests that the communities are aware that there are value addition opportunities that they are currently not realising.

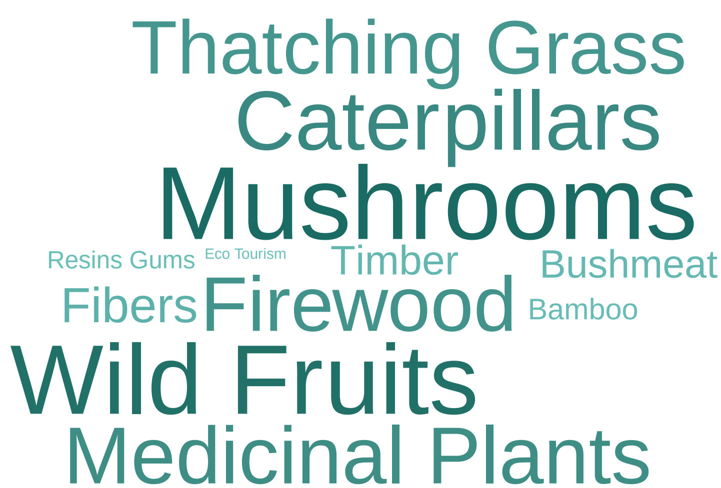

Figure 15: Forest products harvested for household use reported by female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting each product.

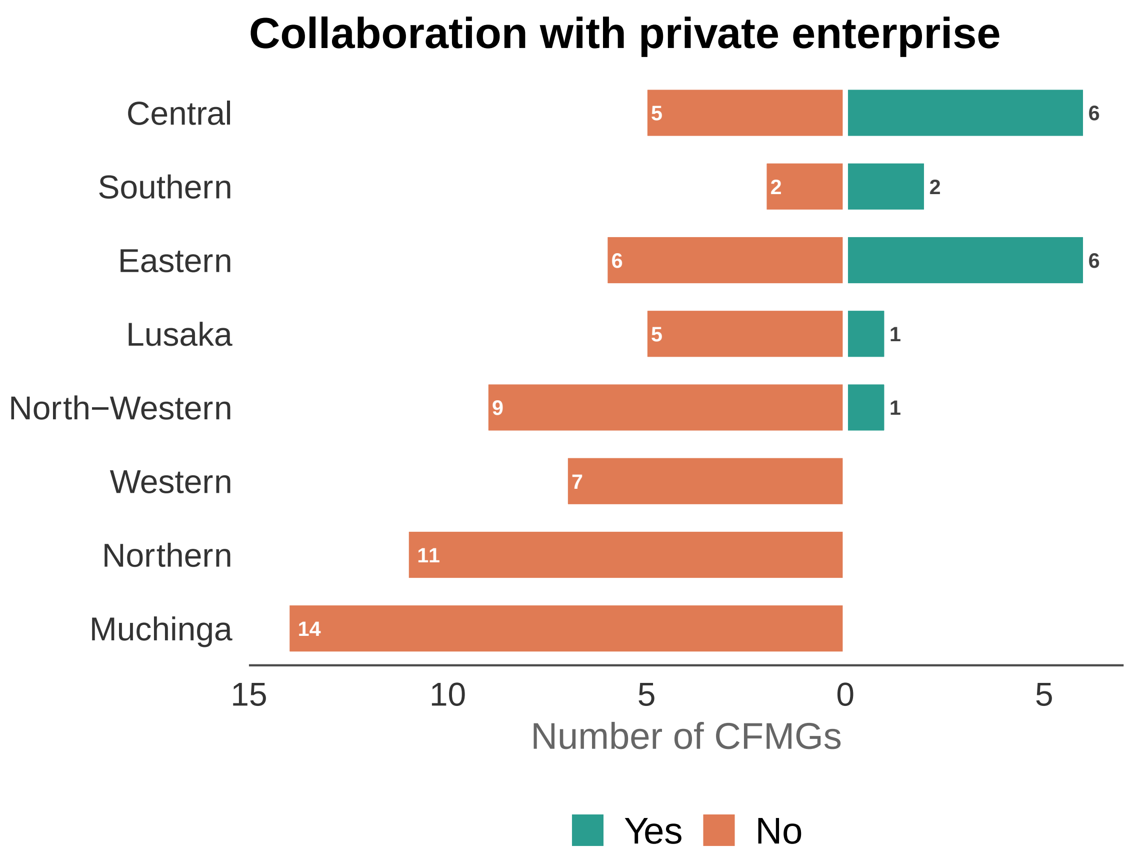

Figure 16: CFMG collaboration with private enterprise as reported by executive FGDs by province.

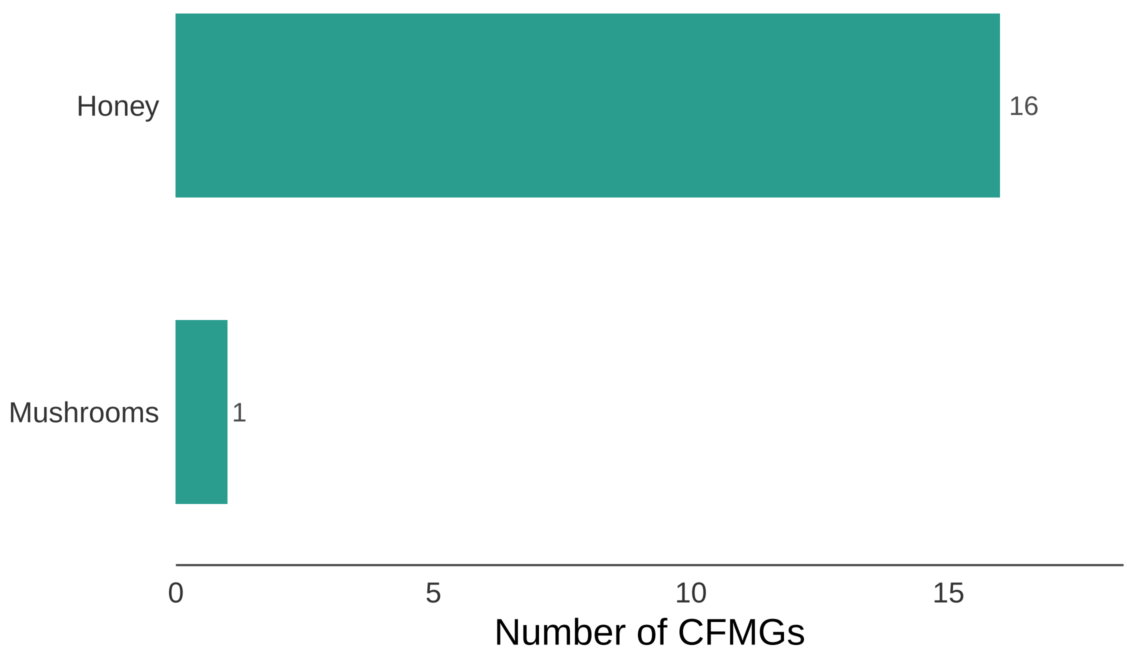

Figure 17: Value chains that CFMGs collaborate with through private enterprise partnerships, as reported by executive FGDs.

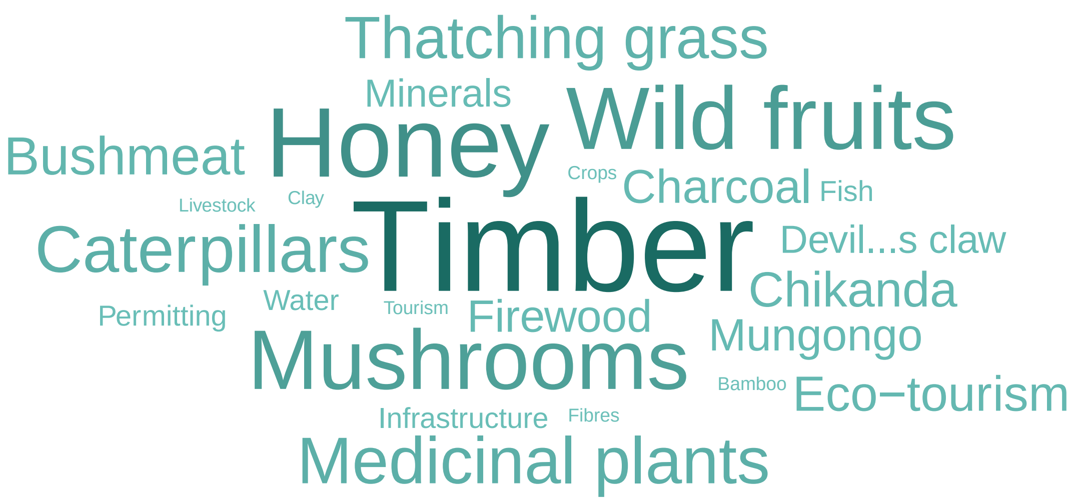

Figure 18: Unrealised economic opportunities identified by FGD groups.

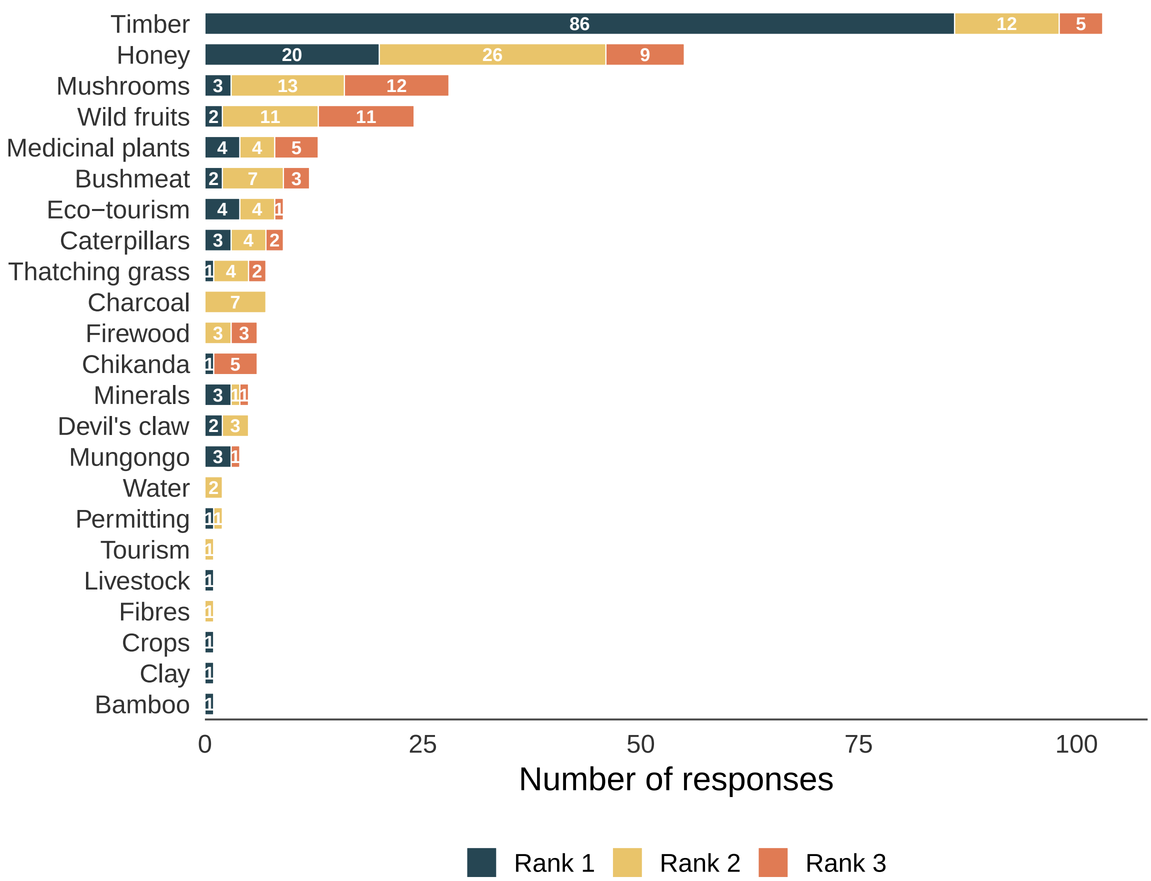

Figure 19: Ranking of unrealised economic opportunities by FGD groups. Bars show the frequency of rank 1 (highest priority), rank 2, and rank 3 responses.

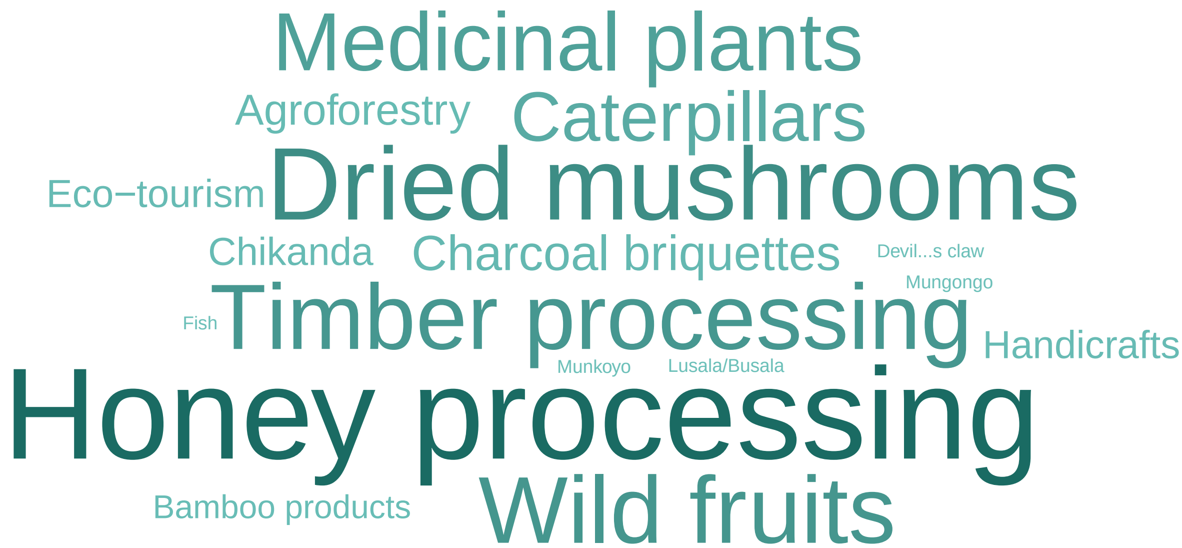

Figure 20: Potential value-added opportunities identified by FGD groups.

We asked the CFMG Executive FGDs to report which primary markets CFM products are sold into (Figure 21). Evidently, most products are sold only at the local market level, with some also being transported to national markets.

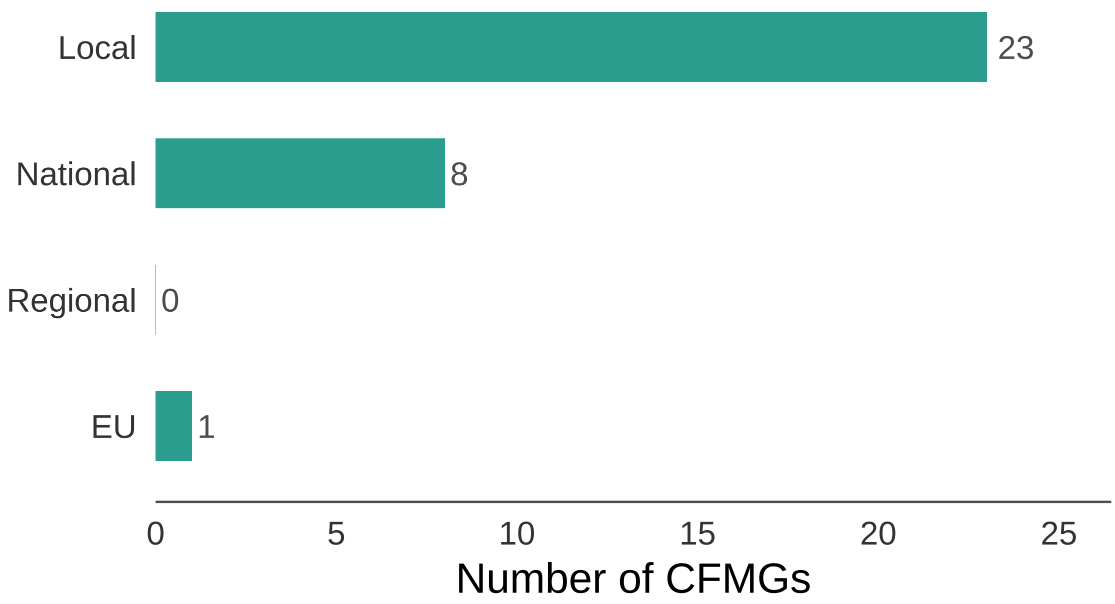

Figure 21: Primary markets for CFM products, as reported by executive FGDs.

Opportunity costs

Precluded activities

The community female, male, and youth FGD groups identified economic activities that are precluded under CFM regulations (Figure 22). While many of these are self evident, the fact that NTFP harvesting is listed quite prominently is surprising. Whether this reflects a misunderstanding on the part of the communities or indeed that they are being told they cannot collect NTFPs is not clear.

Other opportunity costs

FGD groups were asked whether CFM entails opportunity costs for the community (Figure 23). Interestingly, many of the FGDs did not recognise any opportunity costs. However, about a third of the FGDs did. The proportion of FGDs recognising opportunity costs was highest among community male FGDs and lowest among community female FGDs. However, none of the pairwise comparisons were significant.

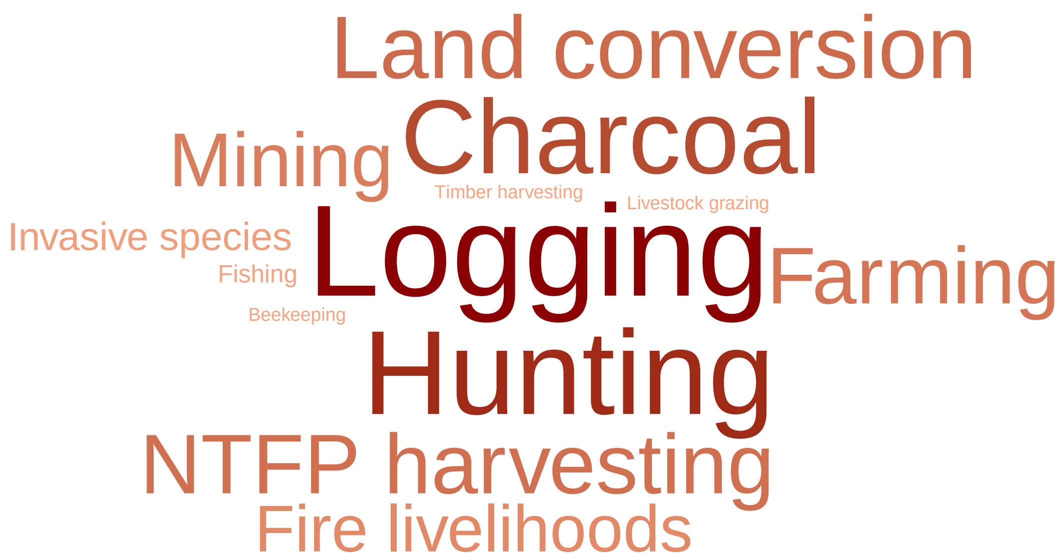

Figure 22: Economic activities precluded by CFM, as reported by FGD groups.

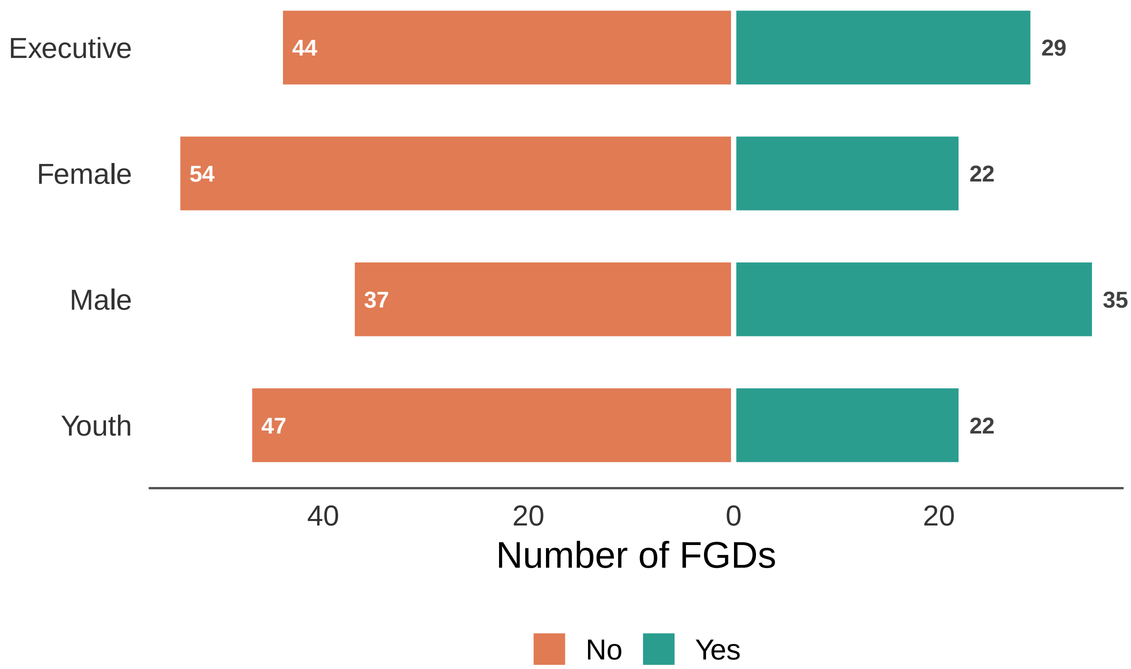

Figure 23: Whether CFM entails opportunity costs for the community, by FGD group.

Table 17: Results of the model examining the effect of FGD type and governance index on whether communities report opportunity costs of CFM.

Model results for opportunity costs of CFM <U+2014> FGD type and governance index

Estimates are log-odds; reference = Executive FGD. Province variance = 0.867; CF-within-province variance = 0.049. Marginal R<U+00B2> = 0.041; conditional R<U+00B2> = 0.25. n = 271.

Factor	Estimate	SE	*P* value
Intercept (Executive FGD)	−0.890	0.542	0.1003
Female FGD	−0.515	0.383	0.1791
Male FGD	0.322	0.374	0.3884
Youth FGD	−0.441	0.385	0.2514
Governance index	0.122	0.072	0.0934

Individual interviewees were also asked whether CFM entails opportunity costs. Table 18 shows the percentage reporting this broken down by interviewee role. Interestingly, the community traditional leaders and community members recognise opportunity costs less of the time than other individual interviewee types, although these differences were not significant.

Table 18: Percentage of individual interviewees reporting that CFM entails opportunity costs for the community, by interviewee role.

Opportunity costs of CFM, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses. A positive response indicates that the respondent considers CFM to entail opportunity costs for the community.

Role	n	Report opportunity costs
Community member	31	47%
CFMG executive member	63	51%
Traditional leader	75	35%
District Forestry Officer	56	39%
NGO representative	32	59%

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Random effect variances not available. Returned R2 does not account for random effects.
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Table 19: Results of the model examining the effect of interviewee role, gender and age on whether individuals report opportunity costs of CFM.

Model results for opportunity costs of CFM <U+2014> individual interviewees

Estimates are log-odds; reference = Community member. Province variance = 0.385; CF-within-province variance = 0. Marginal R<U+00B2> = 0.033; conditional R<U+00B2> = NA. n = 245.

Factor	Estimate	SE	*P* value
Intercept (Community member)	−0.955	0.810	0.2382
CFMG executive member	−0.109	0.482	0.8209
Traditional leader	−0.344	0.477	0.4711
District Forestry Officer	−0.203	0.507	0.6890
NGO representative	0.721	0.576	0.2106
Male	0.405	0.360	0.2599
Age	0.007	0.012	0.5414

The types of opportunity costs identified are shown in Figure 24. Mentions of reduced access, land restrictions and land use conflict are by some margin the most frequently identified opportunity costs. However, it is also interesting that time costs and diverting community resources are also quite prevalent. CFM requires that community members meet which entails a time cost and community resources are diverted to managing the forest that might otherwise be spent on other activities, such as agriculture.

Economic transformation index

To summarise the overall economic transformation performance of each CFMG, we constructed an economic transformation score (0–5) by summing five binary indicators reported by FGD groups — improved access to natural resources for household use, improved access to natural resources for sale, new employment opportunities created, collaboration with a private enterprise (executive FGDs only), and absence of opportunity costs (No = 1, Yes = 0) — scored as 1 (positive outcome reported) or 0 (no positive outcome or worsening). The distribution of economic transformation scores across CFMGs is shown in Figure 25. We then modelled this score using a linear mixed model with FGD type and the governance index as predictors, and random intercepts for province and CF within province, to test whether governance and FGD type were significant predictors of overall economic transformation performance (Table 20, Figure 26).

Table 20: Results of the model examining the effect of FGD type and governance index on economic transformation performance.

Model results for economic transformation performance <U+2014> FGD type and governance index

Linear mixed model with random intercepts for province and CF within province. Reference = Executive. Province variance = 0.02; CF-within-province variance = 0.219; residual variance = 1.054. Marginal R<U+00B2> = 0.094; conditional R<U+00B2> = 0.262. n = 273.

Factor	Estimate	SE	*P* value
Intercept	1.528	0.232	0.0000
Female	−0.256	0.172	0.1396
Male	−0.280	0.176	0.1125
Youth	0.053	0.174	0.7609
Governance index	0.158	0.041	0.0003

Discussion

Through the FGDs and the individual interviews, we investigated a diverse set of indicators of economic transformation. We covered access to natural resources for household use and sale, employment, income earning opportunities, including carbon finance and other sources of income, and opportunity costs.

Most FGD groups and individual interviewees indicated that CFM had not increased access to natural resources for either household use or sale. Given that the communities were involved in harvesting forest products before the instigation of CFM, it is not surprising that most respondents do not recognise any change. However, around one third of respondent groups and individuals felt that access had improved under CFM. This presumably reflects the formalisation of community rights to access forest resources.

A larger proportion of FGD groups considered that CFM had had a positive effect on full-time employment. However, the differences among different FGD types were highly significant. More community female groups suggested there was no change in employment and more community youth groups felt full-time formal employment opportunities had increased. We also showed that the numbers employed included many more men than women and, given that by far the most important employment roles were HFOs and Community Scouts, it is likely they also included a disproportionate number of youth, which would explain these differences in perception. Moreover, while CFM does appear to have increased formal employment, the numbers are quite paltry. In part, this may reflect the fact that employment in forest product harvesting and processing is rarely formal or full-time. While CMF does appear to have resulted in some new employment opportunities, these have been largely limited to HFO and Community Scout roles and have not benefited many women.

A large proportion of CFs in Zambia were formed on the promise of carbon dollars. However, only approximately one third of the CFs in our sample appeared to have benefited from carbon

Figure 24: Types of opportunity costs experienced under CFM.

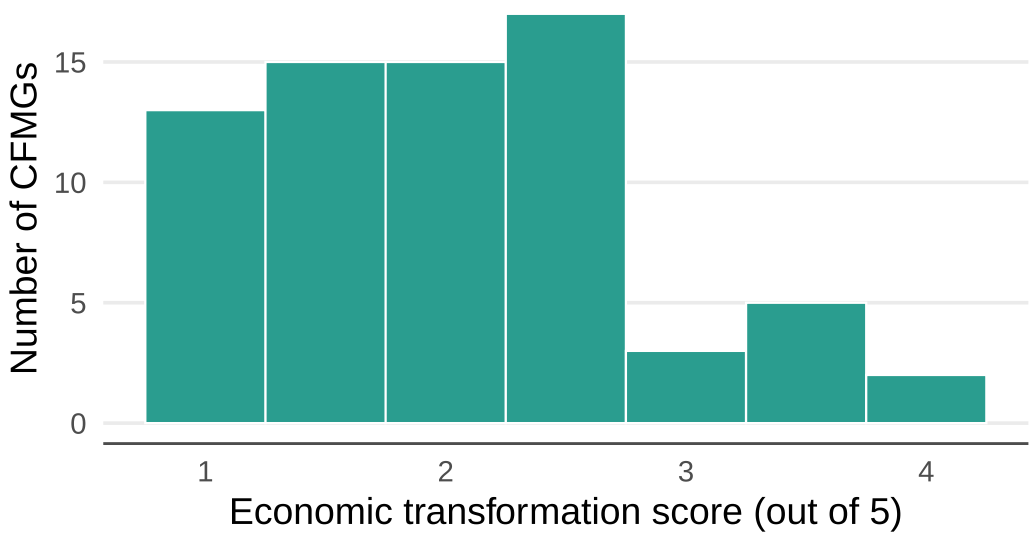

Figure 25: Distribution of mean economic transformation scores across CFMGs (0–5). Each bar represents the number of CFMGs with that mean score, averaged across all FGD types. The score sums five binary indicators: improved household resource access, improved resource access for sale, new employment opportunities, collaboration with a private enterprise, and absence of opportunity costs.

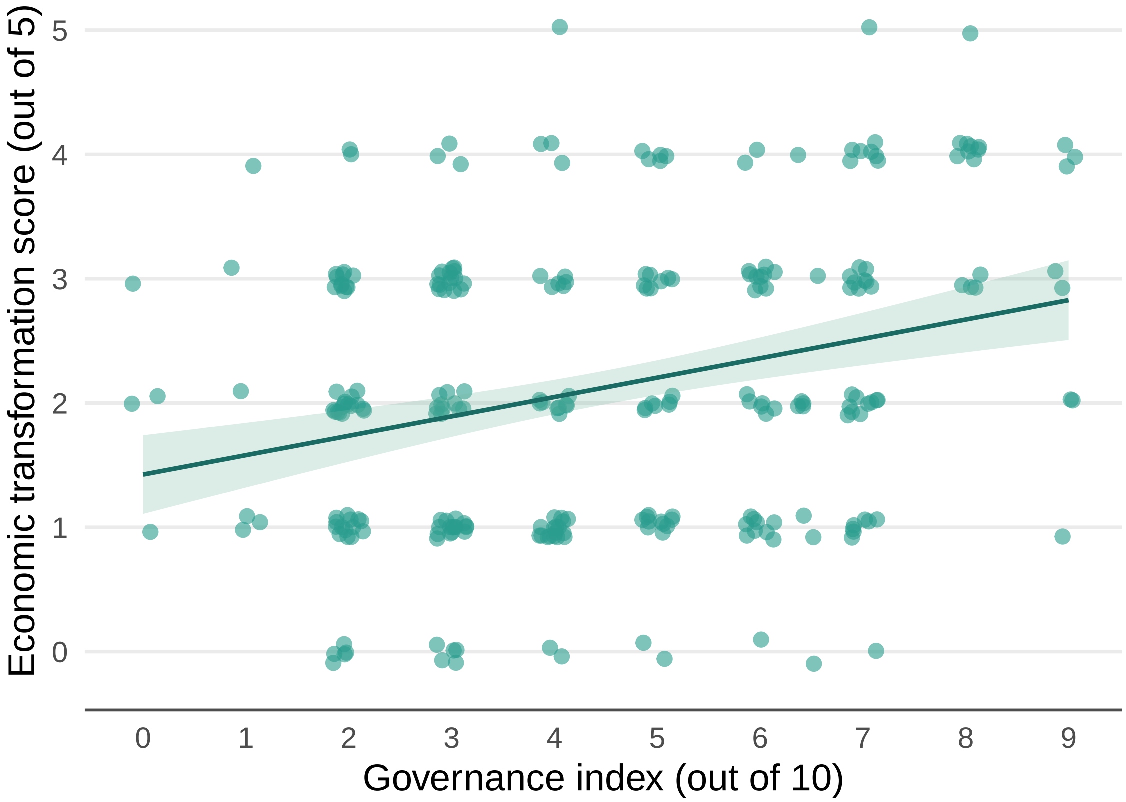

Figure 26: Relationship between the CFMG governance index and economic transformation score. Each point represents one FGD session. The line shows the fitted relationship from the linear mixed model, with the shaded band representing ± 1 SE.

income at the time of the survey. The fact that a number of the FDG groups, especially youth groups, were unaware of the carbon agreement is concerning. It suggests that the carbon trading companies may have failed to conduct a proper FPIC process involving the entire community and that the CFMG executives may not be communicating properly with their communities. Nevertheless, many carbon traders are clearly supporting communities. Most of the activities concern forest protection and agricultural extension for sustainable intensification, for example, climate smart agriculture. Both of which may be considered as essential to a carbon business plan. NTFPs and enterprise development were less often mentioned as a part of the carbon trader activities. This is perhaps an oversight on their part, because enhancing forest product value chains would increase the value of forest conservation for local communities.

The above results are backed up by the response to questions concerning other income earning opportunities. It seems CFM has not led to increased income from market linkages, alternative livelihoods, conservation revenue, forest product processing or ecotourism except for in a very few cases. Nevertheless, a wide variety of forest products clearly contribute to household use and some additional income. Although honey and mushrooms were most often identified as the most important source of additional income, another ten NTFPs were mention as the most important at least once. However, these additional sources of income were identified in less than half of the CFs and only 13 identified a second rank. Thus, forest products are rarely being exploited as a source of additional income and if they are it is usually only one. Only a small number of CFMGs are partnering with an enterprise, such as an off-taker, and among those that do it almost exclusively concerns honey. Nonetheless, the communities identified a wide diversity of forest products that could potentially be harvested and processed to provide an additional source of income. Prominent among these were timber and wild fruit, as well as mushrooms and honey. In some cases timber harvesting may conflict with a carbon business plans, but not if plantations were established. The potential of CFM to support forest product enterprises is clearly massively under realised.

Unfortunately, the information on incomes from CFM were not very informative. Some clearly included income from carbon while most did not and it seems that the CFMG executives have little understanding of the value of forest products being harvested from their CFs. To get a more accurate picture of these values a more in depth PEN type survey is recommended.

The local communities identified a number of opportunity costs to CFM. Most of these pertained to activities that are precluded, such as those involving forest clearance. However, many groups also identified other opportunity costs, such as time costs and diverted resources. Recognition of opportunity costs is important to understand the cost-benefits of CFM for communities, especially as these costs are likely to be unevenly distributed among community members.

Overall, a positive picture of economic transformation emerges from the responses of the FGDs and individual interviews. While changes in indicators, such as employment or additional income from forest products, many not appear substantial, they are in a positive direction. Most CFMGs will also receive carbon income, even if this has only benefited around one third of CFMGs so far. However, what is also evident is that CFMG business plans are over-reliant on just carbon or sometimes carbon and honey. Substantial opportunities for developing much more diversified business plans exist. These could include a range of different NTFPs and potentially also timber. In addition to simply increasing income earning opportunities, diversified business plans would also provide a buffer against fluctuating commodity prices and because different products are harvested at different times of the year, they can help stabilise income throughout the year.